

Period: _____

Date: _____

Objective: Use isolines to represent field data, construct a topographic profile from a contour map, and calculate gradient using the ESRT equation.

1. Mapping the Earth: Terms

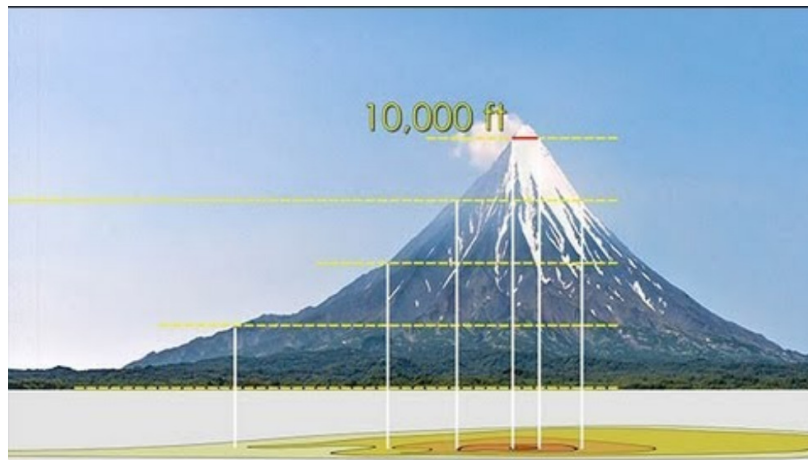
A **field** is any region of space that has a measurable value of a given quantity at _____.

An **isoline** is a line that connects points of _____.

Fill in the table:

Type of isoline	Connects points of equal. . .	Where you see it
Isotherm	_____	Weather map
Isobar	_____	Weather map
Contour line	_____	Topographic map

A topographic map shows a _____ landscape on a _____ sheet of paper. Each contour line traces one elevation all the way around the landform.



2. How to Draw an Isoline

1. Connect dots of _____, using the interval stated in the directions.
2. Where the exact number is not printed, _____ where it would fall between a point of higher value and a point of lower value.
3. Bring every isoline all the way to the _____.

Rules isolines never break:

- Isolines never _____ each other.
- Isolines never split or branch.
- Isolines close into loops or run off the map edge.
- Values change **in order** — no skipping.
- Lines close together = value changing _____.
- Lines far apart = value changing _____.

Practice: elevation field (contour interval = 10 ft)

Draw contour lines at 40, 50, 60, 70, and 80 feet. Label each line and carry it to the edge of the box.

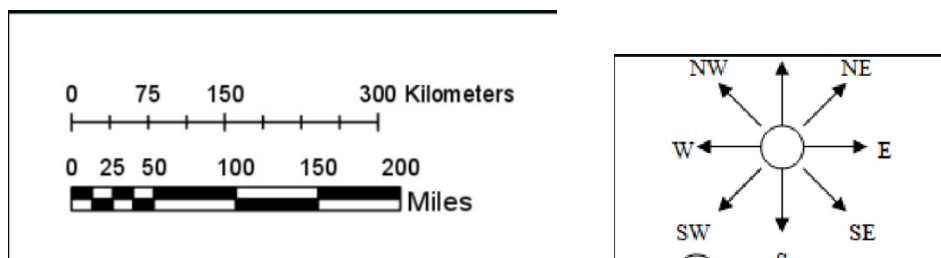
.42	.48	.55	.61	.58	.51
.47	.56	.68	.72	.65	.55
.52	.63	.78	.85	.71	.58
.49	.58	.70	.76	.66	.54
.44	.51	.59	.63	.57	.48
.38	.43	.48	.52	.47	.41

Elevation (feet)

3. Map Scale and Direction

Map scale: a ratio of the distance between two places on a map and the _____ on Earth's surface.

Cardinal directions: _____, _____, _____, _____, and any combination of these (NE, SW, ...).

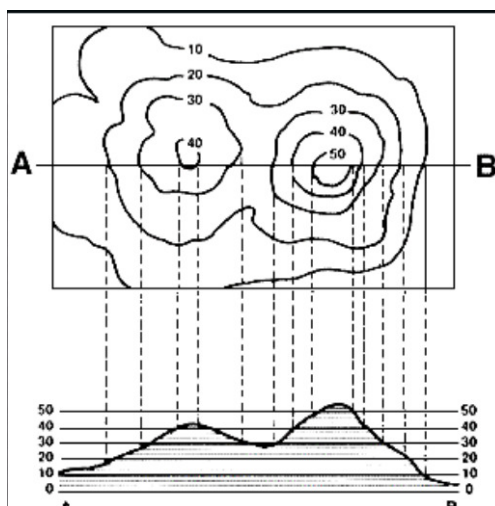


4. Constructing a Profile

A **profile** is a _____ of the land along a line drawn on the map.

1. Line the edge of a piece of _____ up with the two points on your map.
2. Mark the two endpoints, plus every _____ that falls between them.
3. Write the _____ of every mark.
4. Line the paper edge up with the _____ of your graph (first point at the origin) and plot each elevation against the _____.
5. Connect your points with a _____.

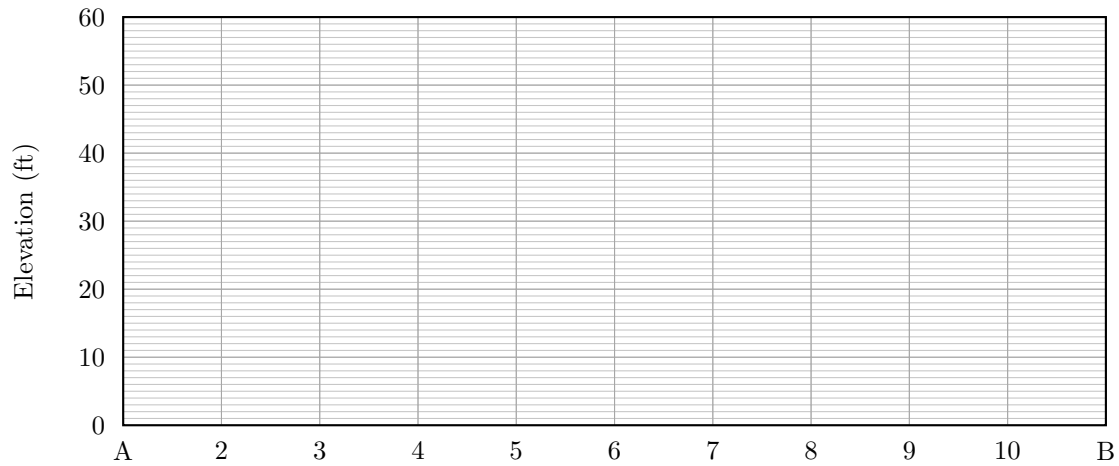
Round out the tops of your peaks and the bottoms of your valleys. The land keeps rising past the last contour line you crossed, so the hilltop is rounded — not a sharp point. **Straight lines and sharp peaks receive no credit.**



Practice: plot the profile

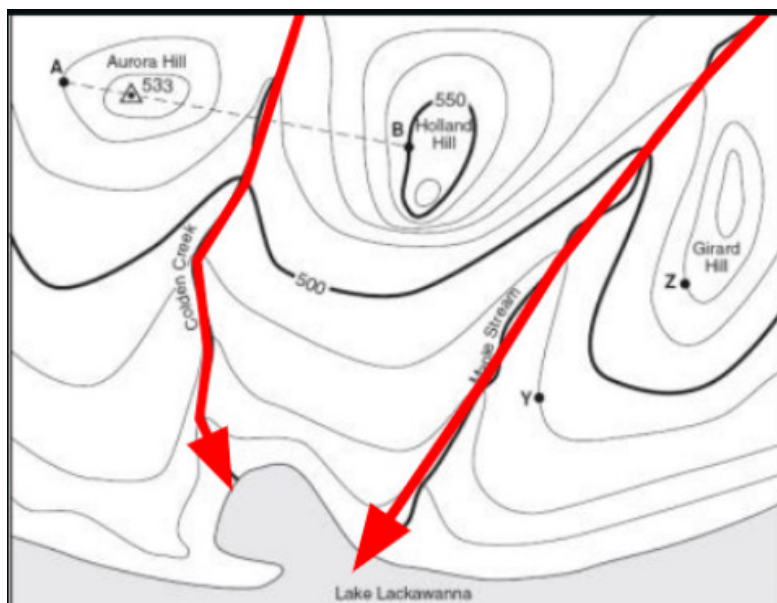
Elevations were read along line A–B and recorded below. Plot each point, then connect them with a smooth, rounded curve.

Point	A	2	3	4	5	6	7	8	9	10	B
Elev. (ft)	20	30	40	40	30	20	30	40	50	40	30



5. Contour Lines and Streams

- Contour lines _____ where they cross a stream.
- The bend forms a V that points _____ (uphill).
- Water flows _____, so the stream flows in the **opposite** direction from the V.
- Streams run from _____ elevation toward _____ elevation.



- (2 points) On the map above, in which compass direction does Colder Creek flow? Explain how the contour lines tell you.

6. Gradient

Gradient tells you how _____ a slope is. The equation is on page _____ of the ESRT:

$$\text{Gradient} = \frac{\text{change in field value}}{\text{distance}}$$

The closer the isolines are together in a given distance, the _____ the gradient.

Worked example. Point A is at 123 feet. Point B is at 20 feet. The points are 2 miles apart.

$$\text{Gradient} = \frac{123 \text{ ft} - 20 \text{ ft}}{2 \text{ mi}} = \frac{\text{ } \text{ft}}{2 \text{ mi}} = \text{ } \text{ft/mi}$$

DON'T FORGET UNITS!

1. (3 points) Point A has an elevation of 260 feet. Point B has an elevation of 10 feet. The points are 3 miles apart. Calculate the gradient. Show the equation, your substitution, and the units.

2. (3 points) A hilltop is 550 meters above sea level. A stream at the base of the hill is 250 meters above sea level. The horizontal distance between them is 1.5 kilometers. Calculate the gradient.

3. (3 points) On a weather map, one station reports 82 °F and another reports 58 °F. The stations are 4 kilometers apart. Calculate the temperature gradient between them.

4. (2 points) Two trails climb the same hill. Trail X crosses contour lines that are crowded close together; trail Y crosses contour lines that are widely spaced. Which trail is steeper, and how do you know?

7. Exit Ticket

1. (1 point) What does a contour line connect?

2. (1 point) A stream crosses a contour line. Which direction does the V point?

3. (1 point) Why must the top of a hill on a profile be drawn rounded instead of pointed?
